

Problem 37. Find all pairs of nonnegative integers (x, y) satisfying

$$2^x = 5^y + 3.$$

Solution 1. For $x \leq 7$, the only solutions (x, y) are $(2, 0)$, $(3, 1)$ and $(7, 3)$.

Suppose that $x \geq 8$. Then $5^y \equiv -3 \pmod{2^8}$. Since $5^3 \equiv 2^7 - 3 \pmod{2^8}$ and the orders of 5 modulo 2^7 and 2^8 are equal to 32 and 64 respectively, we must have $y \equiv 3 \pmod{32}$, but not $y \equiv 3 \pmod{64}$. Hence $y \equiv 35 \pmod{64}$.

Now we need some modulus of the form $64k + 1$, so we take 257. We see that 2^x is congruent to one of

$$\pm 1, \pm 2, \pm 4, \pm 8, \pm 16, \pm 32, \pm 64, \pm 128$$

modulo 257. On the other hand,

$$5^{35} = (5^7)^5 \equiv (-3)^5 \equiv 14 \pmod{257},$$

whereas from $(5^{64k})^4 \equiv 1 \pmod{257}$ we infer that $5^{64k} \in \{\pm 1, \pm 16\}$, so $5^y \in \{\pm 14, \pm 33\}$ and

$$5^y + 3 \in \{-30, -11, 17, 36\} \not\equiv \pm 2^x$$

modulo 257. Thus there are no solutions when $x \geq 8$. □

Solution 2. As in the first solution, we assume that $x \geq 8$. Reduction modulo 2^8 gives us $y \equiv 35 \pmod{64}$. Now we will use modulus 641. This number is convenient because it is well-known that

$$5^4 + 2^4 = 641 \mid 2^{32} + 1.$$

So $5^4 \equiv -2^4 \pmod{641}$. It follows that

$$5^{32} \equiv 2^{32} \equiv -1 \pmod{641}, \quad 5^{64} \equiv 1 \pmod{641} \quad \text{and} \quad 5^y \equiv 5^{35} \equiv -5^3 = -125 \pmod{641}.$$

Hence $2^x = 5^y + 3 \equiv -122 \pmod{641}$.

On the other hand, because $2^y \equiv 3 \pmod{5}$, we find that $x \equiv 3 \pmod{4}$, so

$$2^x \in \{\pm 8, \pm 128, \pm 125, \pm 77, \mp 50, \mp 159, \pm 20, \pm 320\}.$$

Thus there are no solutions when $x \geq 8$. □

Problem 38. Suppose that each positive integer is used once to label each of the vertices of an infinite graph and no vertices are left unlabelled. An edge exists between two vertices labelled by positive integers a and b if and only if

$$a + b + 1 \mid a^2 + b^2 + 1.$$

Is this graph connected?

Solution. For any positive integers a, b , we denote by $a \leftrightarrow b$ if and only if vertices a and b are connected by an edge.

Now, observe that, for any positive integer a , we have

$$a^2 + a + 1 \mid (a^2 - a + 1)(a^2 + a + 1) = a^4 + a^2 + 1,$$

and so $a \leftrightarrow a^2$. Similarly, we have $a + 1 \leftrightarrow (a + 1)^2$.

Next we note that

$$\begin{aligned} (a^2)^2 + ((a + 1)^2)^2 + 1 &= (a^2)^2 + a^2 + 1 + ((a + 1)^2)^2 - a^2 \\ &= (a^2 - a + 1)(a^2 + a + 1) + ((a + 1)^2 - a)((a + 1)^2 + a) \\ &= (a^2 + a + 1)(a^2 + (a + 1)^2 + 1), \end{aligned}$$

which means that $a^2 + (a + 1)^2 + 1 \mid (a^2)^2 + ((a + 1)^2)^2 + 1$ and so $a^2 \leftrightarrow (a + 1)^2$.

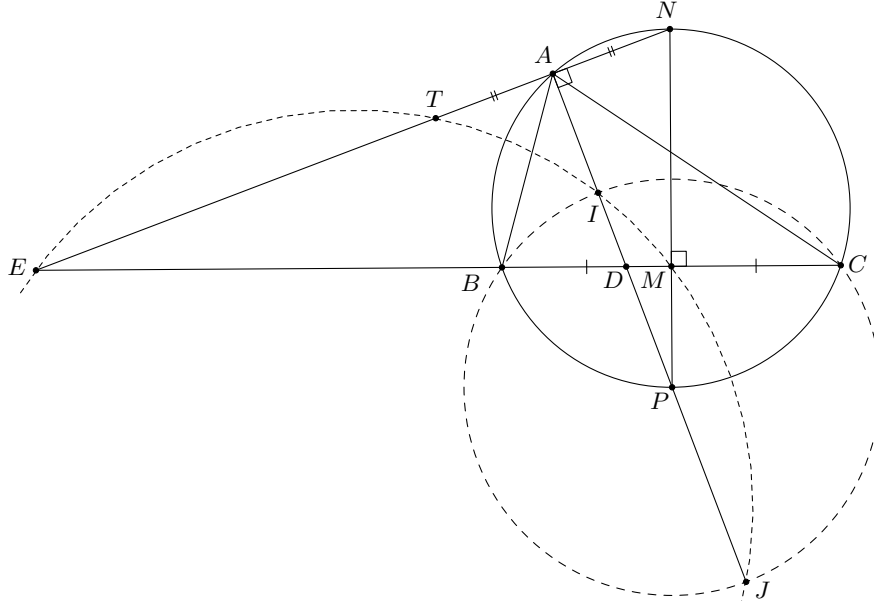
Hence we have

$$a \leftrightarrow a^2 \leftrightarrow (a + 1)^2 \leftrightarrow a + 1,$$

that is, vertices labelled by a and $a + 1$ are connected by an edge. Therefore the graph is connected. □

Problem 39. Let I be the incentre of $\triangle ABC$ and J the excentre opposite vertex A . Denote by M the midpoint of BC and by N the midpoint of $\widehat{BAC}$. If T is the point symmetric to point N with respect to point A , prove that J, M, I, T are concyclic.

Solution 1.



We refer to the configuration above for the proof given below. The other configurations are proved in a similar way. Now suppose that AI cuts BC at D and cuts the circumcircle of $\triangle ABC$ again at P . It is well-known that B, C, I, J lie on circle centred at P with diameter IJ .

We extend AN so that it meets BC at E . Since N is the midpoint of $\widehat{BAC}$, we infer that AE is the external angle bisector of $\angle BAC$. Hence, by the Angle Bisector Theorem, we have $BE \times CD = BD \times CE$. Now observe that

$$\begin{aligned} DE \times DM &= (BE + BD) \left(CD - \frac{1}{2} BC \right) \\ &= BE \times CD + BD \times CD - \frac{1}{2} BC \times DE \\ &= BD \times CE + BD \times CD - \frac{1}{2} BC \times DE \\ &= BD(2 \cdot CD + DE) - \frac{1}{2} BC \times DE \\ &= 2 \cdot BD \times CD - DE \times DM, \end{aligned}$$

and so $DE \times DM = BD \times CD$. Since $BD \times CD = ID \times JD$, we have $DE \times DM = ID \times JD$, and so E, J, M, I are concyclic.

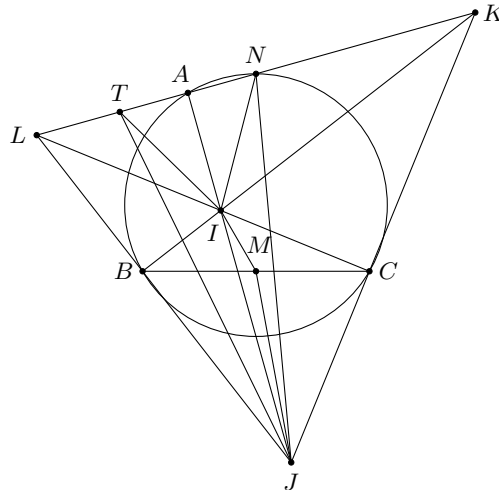
Now observe that $\triangle ABE \sim \triangle ANC$ and $\triangle ABI \sim \triangle AJC$. So $AE \times AN = AB \times AC$ and $AI \times AJ = AB \times AC$. Hence,

$$AT \times AE = AN \times AE = AB \times AC = AI \times AJ,$$

and so E, J, I, T are concyclic.

Therefore, we combine the results above and see that J, M, I, T are concyclic. □

Solution 2.



Let K and L be the B -excentre and C -excentre of $\triangle ABC$. Then N is the midpoint of KL and $\triangle ABC$ is the orthic triangle of $\triangle JKL$. It follows that $\triangle JBC \sim \triangle JKL$ and $\triangle IBC \sim \triangle ILK$. Since M and N are the midpoints of the corresponding sides BC and KL , we have $\triangle JBM \sim \triangle JKN$ and $\triangle IBM \sim \triangle ILN$. Therefore,

$$\angle IMJ = \angle IMB + \angle BMJ = \angle INL + \angle KNJ = 180^\circ - \angle INJ.$$

Also, note that AJ is the perpendicular bisector of NT . Thus, $\angle INJ = \angle ITJ$. It follows from above that $\angle IMJ = 180^\circ - \angle ITJ$. Hence, J, M, I, T are concyclic. $\square$

Problem 40. Let $\{x_n\}$ and $\{y_n\}$ be sequences of real numbers satisfying $x_1 = y_1 = 1$,

$$x_{n+1} = \frac{x_n + 2}{x_n + 1} \quad \text{and} \quad y_{n+1} = \frac{y_n^2 + 2}{2y_n} \quad \text{for } n = 1, 2, \dots$$

Prove that $y_{n+1} = x_{2^n}$ for $n = 0, 1, 2, \dots$

Solution 1. First of all, we note that, for $n \geq 2$,

$$x_{n+1} = \frac{x_n \cdot x_1 + 2}{x_n \cdot x_1 + 1} = \frac{\frac{x_{n-1} + 2}{x_{n-1} + 1} \cdot x_1 + 2}{\frac{x_{n-1} + 2}{x_{n-1} + 1} + x_1} = \frac{x_{n-1}(x_1 + 2) + 2(x_1 + 1)}{x_{n-1}(x_1 + 1) + (x_1 + 2)} = \frac{x_{n-1} \cdot \frac{x_1 + 2}{x_1 + 1} + 2}{x_{n-1} + \frac{x_1 + 2}{x_1 + 1}} = \frac{x_{n-1} \cdot x_2 + 2}{x_{n-1} + x_2},$$

and so, inductively, we obtain

$$x_n = \frac{x_k x_{n-k} + 2}{x_k + x_{n-k}} \quad \text{for } k = 1, \dots, n-1.$$

Next, we let $a_{n+1} = x_{2^n}$ for $n = 0, 1, 2, \dots$. Then, choosing $k = 2^{n-1}$, we have

$$a_{n+1} = x_{2^n} = \frac{x_{2^{n-1}} x_{2^{n-1}} + 2}{x_{2^{n-1}} + x_{2^{n-1}}} = \frac{x_{2^{n-1}}^2 + 2}{2x_{2^{n-1}}} = \frac{a_n^2 + 2}{2a_n}.$$

Again, by induction, $a_{n+1} = y_{n+1}$ for $n = 0, 1, 2, \dots$ and the result follows. $\square$

Solution 2. The first few terms of $\{x_n\}$ are $\frac{1}{1}, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \dots$. The numerators and the denominators are just the solutions to $a^2 - 2b^2 = \pm 1$. Inspired by this, we define

$$a_n + b_n \sqrt{2} = (1 + \sqrt{2})^n$$

where a_n, b_n are integers. It is easy to deduce $a_{n+1} = a_n + 2b_n$ and $b_{n+1} = a_n + b_n$. Therefore, if $x_n = \frac{a_n}{b_n}$, then

$$x_{n+1} = \frac{x_n + 2}{x_n + 1} = \frac{a_n + 2b_n}{a_n + b_n} = \frac{a_{n+1}}{b_{n+1}}.$$

Since $x_1 = \frac{a_1}{b_1}$, we can prove by induction that $x_n = \frac{a_n}{b_n}$ for any $n \in \mathbb{N}$.

Similarly, define

$$c_n + d_n \sqrt{2} = (1 + \sqrt{2})^{2^{n-1}}$$

where c_n, d_n are integers. From $c_{n+1} + d_{n+1} \sqrt{2} = (c_n + d_n \sqrt{2})^2$, we deduce $c_{n+1} = c_n^2 + 2d_n^2$ and $d_{n+1} = 2c_n d_n$. Therefore, if $y_n = \frac{c_n}{d_n}$, then

$$y_{n+1} = \frac{y_n^2 + 2}{2y_n} = \frac{c_n^2 + 2d_n^2}{2c_n d_n} = \frac{c_{n+1}}{d_{n+1}}.$$

Since $y_1 = \frac{c_1}{d_1}$, we can prove by induction that $y_n = \frac{c_n}{d_n}$ for any $n \in \mathbb{N}$. By definition, it is obvious that $c_{n+1} + d_{n+1} \sqrt{2} = a_{2^n} + b_{2^n} \sqrt{2}$. So

$$y_{n+1} = \frac{c_{n+1}}{d_{n+1}} = \frac{a_{2^n}}{b_{2^n}} = x_{2^n}.$$

$\square$